



Trigonometry Word Problems Using SOH CAH TOA

Trigonometry Worksheet · Grade 9-11 · numberbender.com

Name: _____

Date: _____

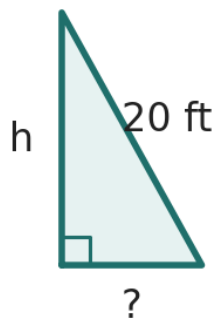
Score: / 10

Learning Objectives

- Identify which trig ratio (sine, cosine, or tangent) applies to a given right-triangle word problem
- Set up and solve equations using SOH CAH TOA to find missing sides or angles
- Apply right-triangle trigonometry to real-life situations involving angles of elevation and depression

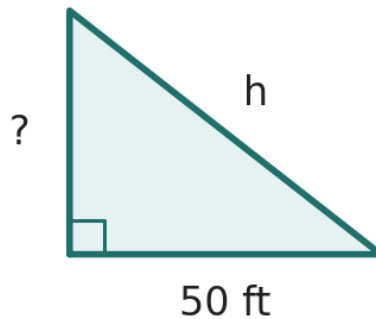
Use SOH CAH TOA to solve each right-triangle word problem; round final answers to the nearest tenth.

1. A 20 ft ladder leans against a wall making a 65° angle with the ground. How high up the wall does the ladder reach?



Answer: _____

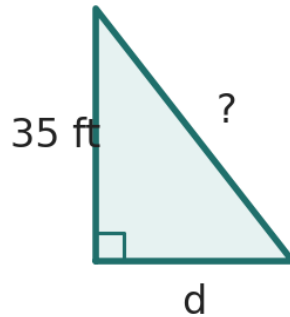
2. From a point on level ground 50 ft from the base of a flagpole, the angle of elevation to the top of the pole is 38° . Find the height of the flagpole.



Answer: _____

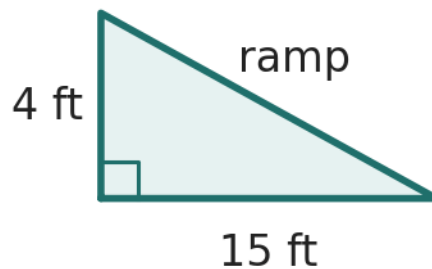


3. A guy wire is attached to the top of a 35 ft tower and anchored to the ground at an angle of 52° with the ground. How long is the guy wire?



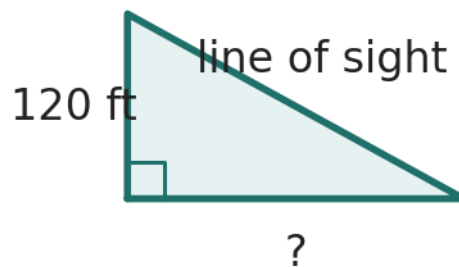
Answer: _____

4. A ramp rises 4 ft over a horizontal distance of 15 ft. Find the angle the ramp makes with the ground.



Answer: _____

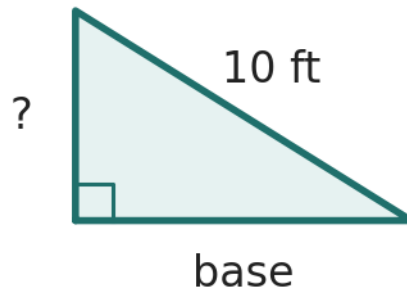
5. From the top of a 120 ft cliff, the angle of depression to a boat at sea is 25° . How far is the boat from the base of the cliff?



Answer: _____

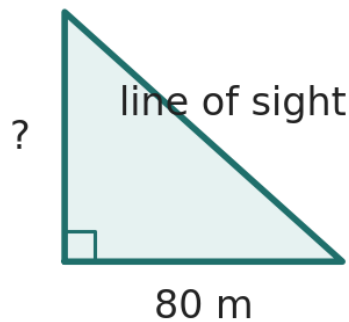


6. A 10 ft slide makes a 32° angle with the ground. How tall is the platform at the top of the slide?



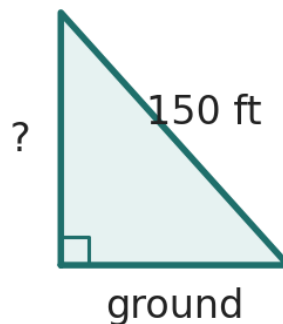
Answer: _____

7. An observer stands 80 m from a building. The angle of elevation to the top of the building is 42° . Find the height of the building.



Answer: _____

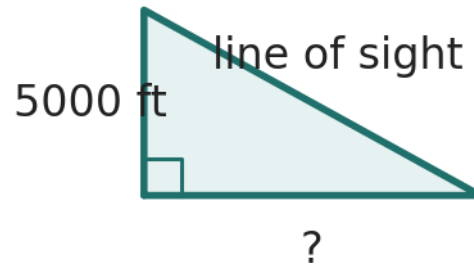
8. A kite string is 150 ft long and makes a 48° angle with the ground. How high is the kite above the ground (assuming the string is straight)?



Answer: _____

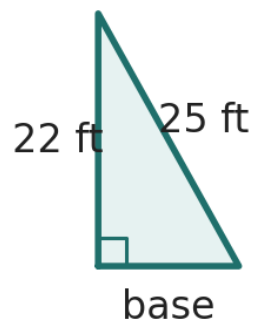


9. A plane is flying at an altitude of 5,000 ft. The pilot sees an airport at an angle of depression of 15° . Find the horizontal distance from the plane to the airport.



Answer: _____

10. A 25 ft ladder reaches 22 ft up a wall. Find the angle the ladder makes with the ground.



Answer: _____

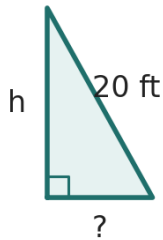




Encourage students to sketch the right triangle, label the opposite/adjacent/hypotenuse relative to the given angle, then choose the correct ratio.

Solutions

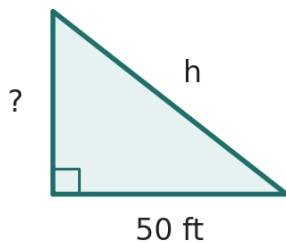
1. A 20 ft ladder leans against a wall making a 65° angle with the ground. How high up the wall does the ladder reach?



- The wall side is opposite the 65° angle and the ladder is the hypotenuse, so use sine.
- Write $\sin(65^\circ) = h / 20$, giving $h = 20 \cdot \sin(65^\circ)$.
- Compute $h \approx 20 \cdot 0.9063 \approx 18.1$ ft.

Answer: ≈ 18.1 ft

2. From a point on level ground 50 ft from the base of a flagpole, the angle of elevation to the top of the pole is 38° . Find the height of the flagpole.

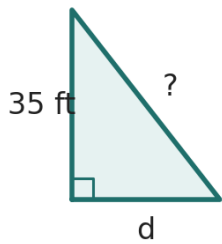


- The 50 ft distance is adjacent to the 38° angle and the height is opposite, so use tangent.
- Write $\tan(38^\circ) = h / 50$, giving $h = 50 \cdot \tan(38^\circ)$.
- Compute $h \approx 50 \cdot 0.7813 \approx 39.1$ ft.

Answer: ≈ 39.1 ft



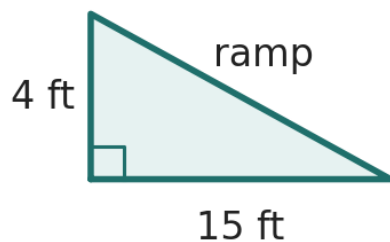
3. A guy wire is attached to the top of a 35 ft tower and anchored to the ground at an angle of 52° with the ground. How long is the guy wire?



- The 35 ft tower is opposite the 52° angle and the wire is the hypotenuse, so use sine.
- Write $\sin(52^\circ) = 35 / L$, giving $L = 35 / \sin(52^\circ)$.
- Compute $L \approx 35 / 0.7880 \approx 44.4$ ft.

Answer: ≈ 44.4 ft

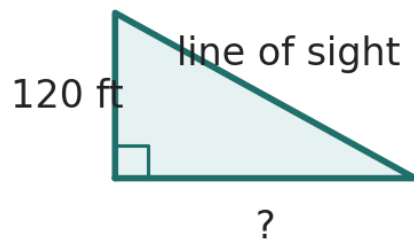
4. A ramp rises 4 ft over a horizontal distance of 15 ft. Find the angle the ramp makes with the ground.



- The rise (4 ft) is opposite the angle and the run (15 ft) is adjacent, so use tangent.
- Write $\tan(\theta) = 4/15$, giving $\theta = \arctan(4/15)$.
- Compute $\theta \approx \arctan(0.2667) \approx 14.9^\circ$.

Answer: $\theta \approx 14.9^\circ$

5. From the top of a 120 ft cliff, the angle of depression to a boat at sea is 25° . How far is the boat from the base of the cliff?

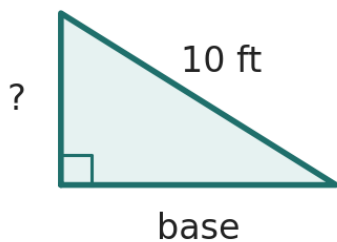


- The angle of depression equals the angle of elevation from the boat, so the 25° angle has the 120 ft cliff opposite and the horizontal distance adjacent.
- Write $\tan(25^\circ) = 120 / d$, giving $d = 120 / \tan(25^\circ)$.
- Compute $d \approx 120 / 0.4663 \approx 257.3$ ft.

Answer: ≈ 257.3 ft



6. A 10 ft slide makes a 32° angle with the ground. How tall is the platform at the top of the slide?



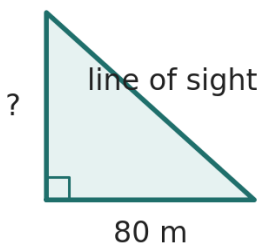
→ The platform height is opposite the 32° angle and the slide is the hypotenuse, so use sine.

→ Write $\sin(32^\circ) = h / 10$, giving $h = 10 \cdot \sin(32^\circ)$.

→ Compute $h \approx 10 \cdot 0.5299 \approx 5.3$ ft.

Answer: ≈ 5.3 ft

7. An observer stands 80 m from a building. The angle of elevation to the top of the building is 42° . Find the height of the building.



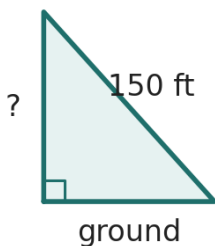
→ The 80 m distance is adjacent to the 42° angle and the height is opposite, so use tangent.

→ Write $\tan(42^\circ) = h / 80$, giving $h = 80 \cdot \tan(42^\circ)$.

→ Compute $h \approx 80 \cdot 0.9004 \approx 72.0$ m.

Answer: ≈ 72.0 m

8. A kite string is 150 ft long and makes a 48° angle with the ground. How high is the kite above the ground (assuming the string is straight)?



→ The height is opposite the 48° angle and the string is the hypotenuse, so use sine.

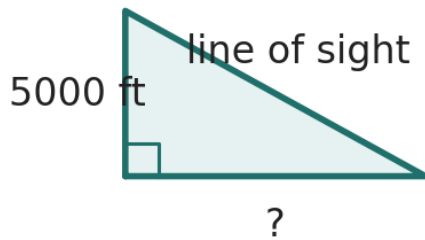
→ Write $\sin(48^\circ) = h / 150$, giving $h = 150 \cdot \sin(48^\circ)$.

→ Compute $h \approx 150 \cdot 0.7431 \approx 111.5$ ft.

Answer: ≈ 111.5 ft



9. A plane is flying at an altitude of 5,000 ft. The pilot sees an airport at an angle of depression of 15° . Find the horizontal distance from the plane to the airport.



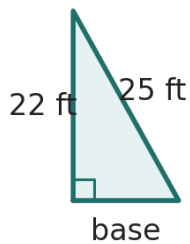
→ The 5,000 ft altitude is opposite the 15° angle (equal to the angle of elevation from the airport) and the horizontal distance is adjacent, so use tangent.

→ Write $\tan(15^\circ) = 5000 / d$, giving $d = 5000 / \tan(15^\circ)$.

→ Compute $d \approx 5000 / 0.2679 \approx 18,660$ ft.

Answer: $\approx 18\,660$ ft

10. A 25 ft ladder reaches 22 ft up a wall. Find the angle the ladder makes with the ground.



→ The 22 ft height is opposite the angle and the 25 ft ladder is the hypotenuse, so use sine.

→ Write $\sin(\theta) = 22/25$, giving $\theta = \arcsin(22/25)$.

→ Compute $\theta \approx \arcsin(0.88) \approx 61.6^\circ$.

Answer: $\theta \approx 61.6^\circ$

